Create the Chen Notation ERD for the following:

Exercise 1 — Customer and Order

An online store maintains information about **customers** and their **orders**.

Each customer has a unique CustomerID, Name, Email, and multiple PhoneNumber. Each order has a unique OrderID, OrderDate, and TotalAmount.

Think about: Is CUSTOMER required to participate in the relationship? Is ORDER required to participate?

Answer:

Assuming a customer can register without placing an order, and every order must belong to a customer:

- **CUSTOMER has optional (partial) participation:** A customer can have **zero or many orders**.
- **ORDER has mandatory (total) participation:** Each order must belong to **exactly one customer**.

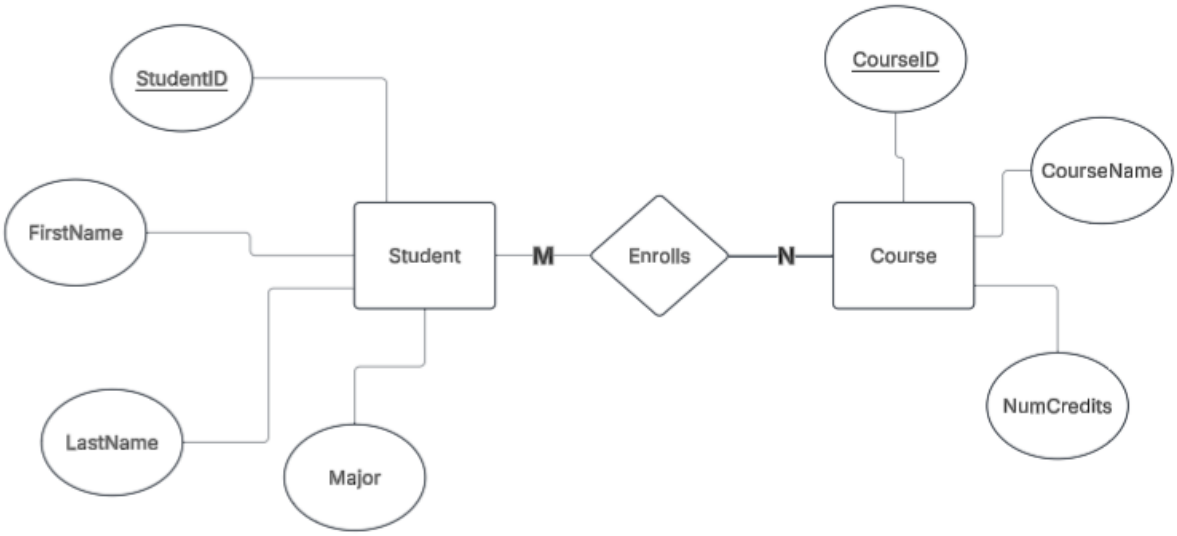
This is a **one-to-many relationship** between CUSTOMER and ORDER.

Exercise 2 — Student and Course

A university maintains information about **students** and **courses**.

Each student has a unique StudentID, FirstName, LastName, and Major. Each course has a unique CourseID, CourseName, and NumberOfCredits.

Answer:



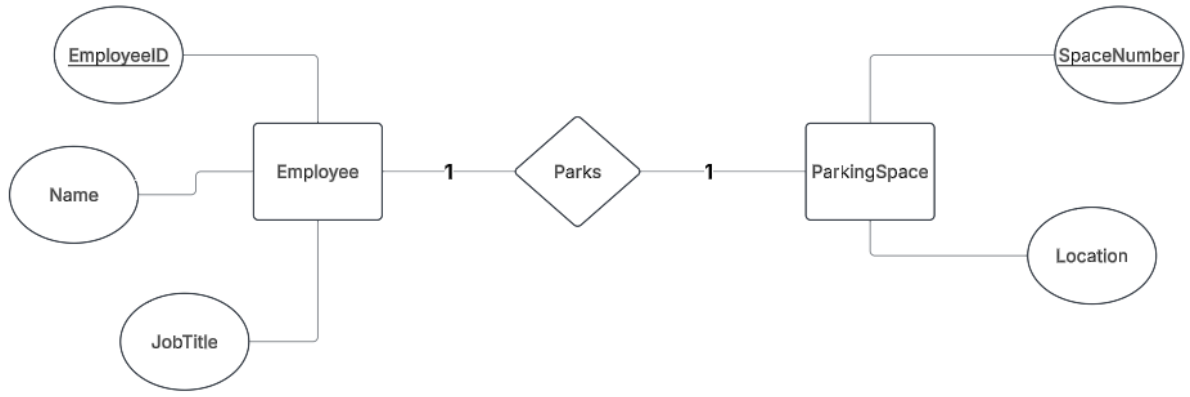
Exercise 3 — Employee and Parking Space

A company provides reserved parking spaces to some employees.

Each employee has an EmployeeID, Name, and JobTitle. Each parking space has a unique SpaceNumber and Location.

An employee may be assigned **zero or one parking space**. A parking space may be assigned to **zero or one employee**.

Answer:



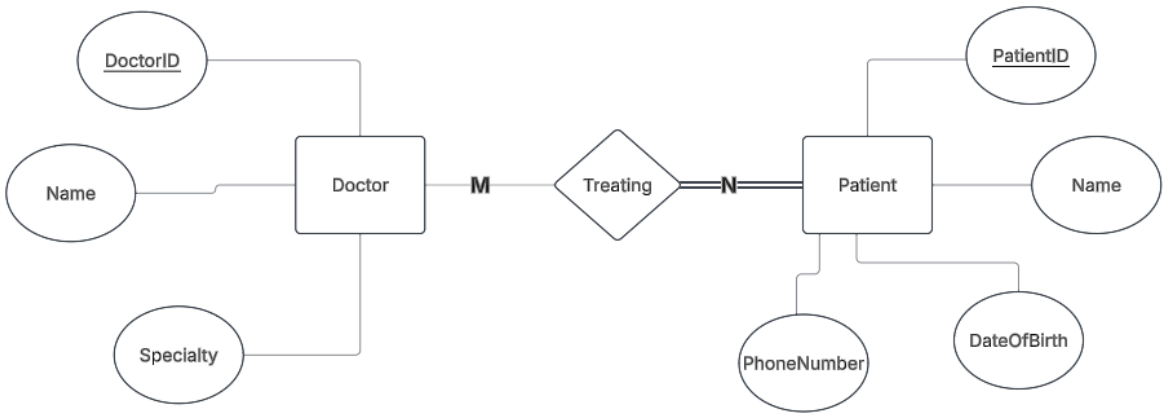
Exercise 4 — Doctor and Patient

A medical clinic keeps records about its **doctors** and **patients**.

Each doctor has a DoctorID, Name, and Specialty. Each patient has a PatientID, Name, DateOfBirth, and PhoneNumber.

A doctor may treat **many patients**, and a patient may be treated by **many doctors**. However, every patient in the database must currently be receiving treatment from **at least one doctor**. A newly hired doctor may not yet have any patients.

Answer:



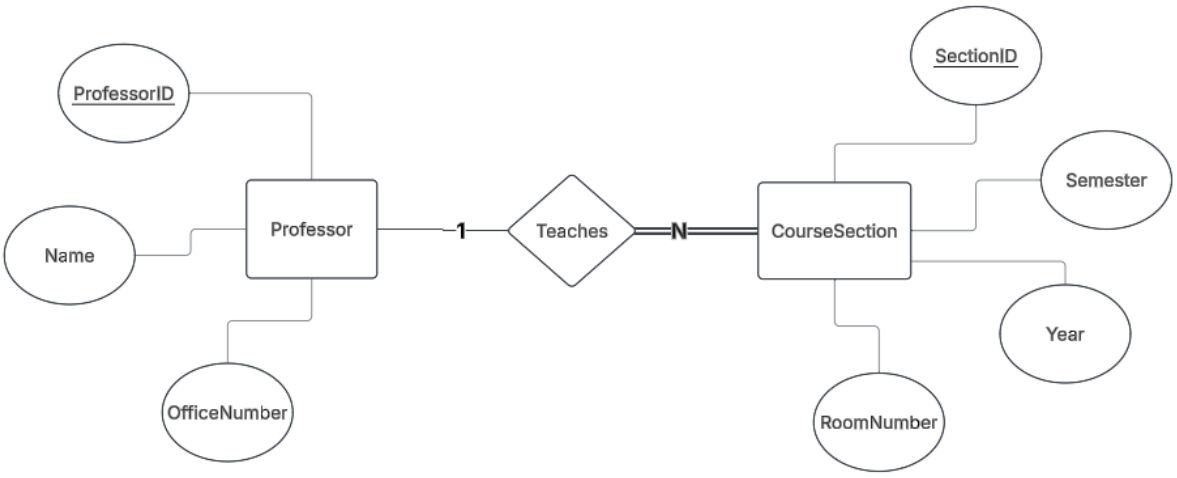
Exercise 5 — Professor and Course Section

A university maintains information about **professors** and **course sections**.

Each professor has a ProfessorID, Name, and OfficeNumber. Each course section has a SectionID, Semester, Year, and RoomNumber.

Every course section must be taught by exactly **one professor**. A professor may teach **zero or many course sections**.

Answer:



Exercise 6 — Customer and Reservation with a Relationship Attribute

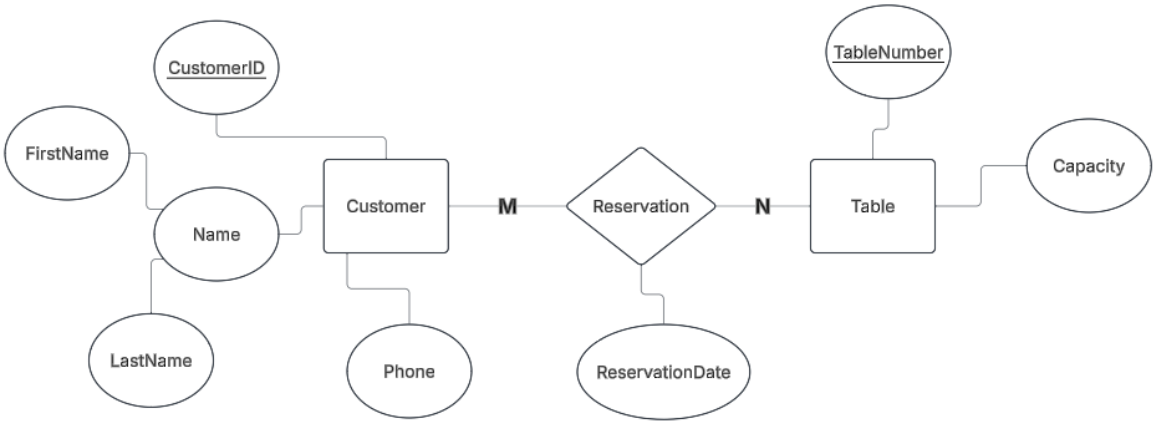
A restaurant records **customers** and **tables**.

Each customer has a CustomerID, Name (Name is a composite attribute), and PhoneNumber. Each table has a TableNumber and Capacity.

A customer may reserve **many tables**, and a table may be reserved by **many customers over time**. A customer does not have to make a reservation, and a table may never have been reserved.

For each reservation, the restaurant records the **ReservationDate**.

Answer:



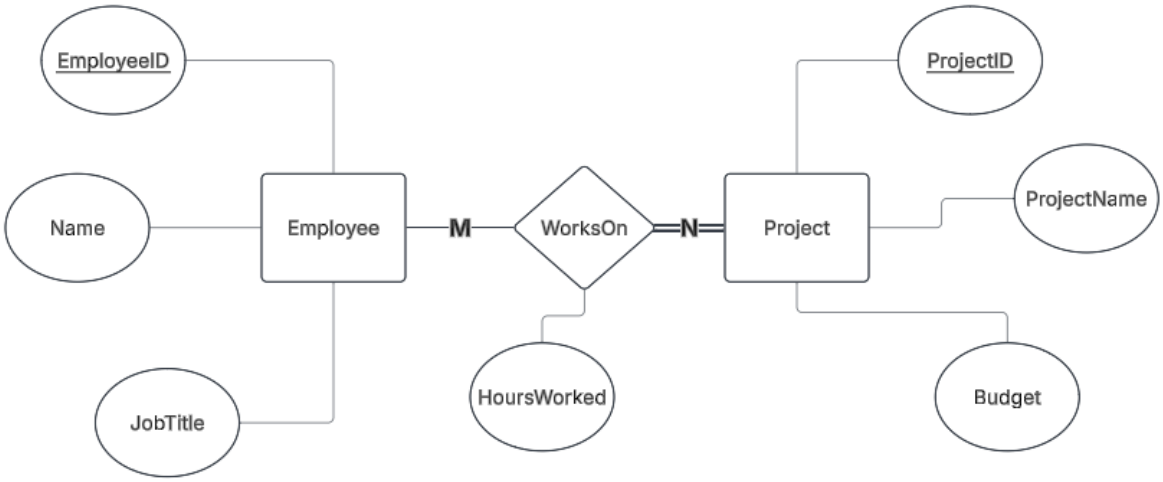
Exercise 7 — Employee and Project

A consulting company assigns employees to projects.

Each employee has an EmployeeID, Name, and JobTitle. Each project has a ProjectID, ProjectName, and Budget.

An employee may work on **zero or many projects**. Every project must have **at least one employee assigned to it and can have many employees**. For each employee's assignment to a project, the company records the **number of HoursWorked**.

Answer:



Exercise 8 — Course pre-requisites

A university maintains information about the **courses** it offers.

Each course has:

- **CourseID**
- **CourseName**
- **CreditHours**
- **Department**

Some courses require students to complete another course before they can enroll.

The following business rules apply:

- A course may have **zero or many prerequisite courses**.
- A course may serve as a **prerequisite for zero or many other courses**.
- Some courses have no prerequisites.
- Some courses are not prerequisites for any other course.

Answer:

